

2 DEFORMATION AND STRAIN

The study of deformation and strain provides the foundation for understanding how materials and structures respond to external actions. In continuum mechanics, deformation describes how the positions of material points change relative to a reference configuration, whereas strain quantifies the associated changes in length, angle, area, and volume. Different strain measures—from infinitesimal approximations to finite-deformation measures—are appropriate under different kinematic assumptions.

2.1 Motion of a continuum

For a single particle P , its path line $\mathbf{x}(t)$ is the trajectory followed by the particle as a function of time t . In continuum mechanics, a particle is identified by the position $\mathbf{X}$ that it occupies in a chosen reference configuration, usually at time t_0 . The motion of the continuum is then described by

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t) \quad (2.1)$$

where $\mathbf{x} = x_1\hat{\mathbf{e}}_1 + x_2\hat{\mathbf{e}}_2 + x_3\hat{\mathbf{e}}_3$ is the current position of the particle that occupied $\mathbf{X} = X_1\hat{\mathbf{e}}_1 + X_2\hat{\mathbf{e}}_2 + X_3\hat{\mathbf{e}}_3$ in the reference configuration. The components of $\mathbf{X}$ are the **material coordinates** and label the particle, whereas the components of $\mathbf{x}$ are its **spatial coordinates** at time t . Figure 2.1 illustrates this mapping between configurations.

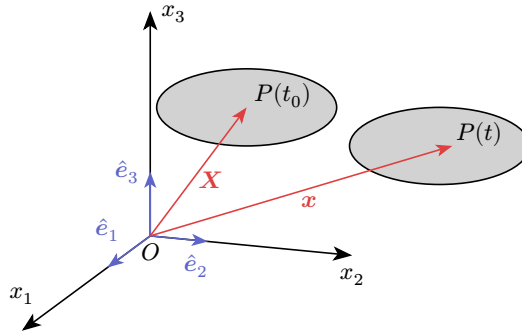
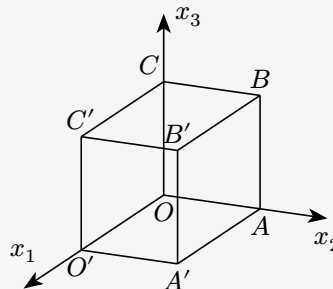


Figure 2.1: Motion of a material point P from the reference to the current configuration.

Example 2.1: Considering the motion

$$\mathbf{x} = \mathbf{X} + ktX_3\hat{\mathbf{e}}_2$$

where k is a constant, draw the configuration at time t of a body that is a unit cube at $t = 0$, as shown in the figure.



Solution:

Expanding the motion equation gives

$$\mathbf{x} = X_1 \hat{\mathbf{e}}_1 + X_2 \hat{\mathbf{e}}_2 + X_3 \hat{\mathbf{e}}_3 + ktX_3 \hat{\mathbf{e}}_2$$

Point O has material coordinates $(X_1, X_2, X_3) = (0, 0, 0)$, so

$$\mathbf{x} = 0\hat{\mathbf{e}}_1 + 0\hat{\mathbf{e}}_2 + 0\hat{\mathbf{e}}_3 + kt0\hat{\mathbf{e}}_2$$

Thus, O remains at the origin for all t . For point A , $(X_1, X_2, X_3) = (0, 1, 0)$, and therefore

$$\mathbf{x} = 0\hat{\mathbf{e}}_1 + 1\hat{\mathbf{e}}_2 + 0\hat{\mathbf{e}}_3 + kt0\hat{\mathbf{e}}_2$$

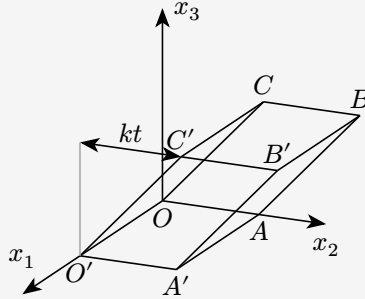
which gives $(x_1, x_2, x_3) = (0, 1, 0)$; hence, A also remains fixed. For point B , $(X_1, X_2, X_3) = (0, 1, 1)$, so

$$\mathbf{x} = 0\hat{\mathbf{e}}_1 + 1\hat{\mathbf{e}}_2 + 1\hat{\mathbf{e}}_3 + kt1\hat{\mathbf{e}}_2$$

which gives $(x_1, x_2, x_3) = (0, 1 + kt, 1)$. For point C , $(X_1, X_2, X_3) = (0, 0, 1)$, and

$$\mathbf{x} = 0\hat{\mathbf{e}}_1 + 0\hat{\mathbf{e}}_2 + 1\hat{\mathbf{e}}_3 + kt1\hat{\mathbf{e}}_2$$

which gives $(x_1, x_2, x_3) = (0, kt, 1)$. The same calculation applies to O' , A' , B' , and C' . Thus, every plane of constant X_3 is translated by ktX_3 in the x_2 direction: the cube undergoes simple shear, as shown below.



2.2 Material and spatial descriptions

For a body in motion, a scalar, vector, or tensor field f may vary with time. The same physical field can be described from two complementary perspectives:

Following the particles: The field is expressed in terms of the material coordinates $\mathbf{X}$ and time t :

$$f = f(X_1, X_2, X_3, t) = f(\mathbf{X}, t) \quad (2.2)$$

This representation is the **material description**, also called the **Lagrangian** or **reference description**. Because $\mathbf{X}$ labels a particle, holding $\mathbf{X}$ fixed means following that particle.

Observing fixed locations: The field is expressed in terms of the spatial coordinates $\mathbf{x}$ and time t :

$$f = f(x_1, x_2, x_3, t) = f(\mathbf{x}, t) \quad (2.3)$$

Material derivative

This representation is the **spatial** or **Eulerian description**. Holding $\mathbf{x}$ fixed describes how the field changes at a fixed location; it does not, by itself, follow the history of a particular particle.

Example 2.2: Given the motion of a body

$$x_1 = X_1 + t, \quad x_2 = X_2 + 2t, \quad x_3 = X_3$$

determine the material description of the spatial field $f = x_1^2 + x_2^2 + x_3^2$. Then find the rate experienced by a material particle and express it in both material and spatial coordinates.

Solution:

Substituting the motion into the spatial field gives its material description:

$$\begin{aligned} f &= f(X_1, X_2, X_3, t) \\ &= (X_1 + t)^2 + (X_2 + 2t)^2 + X_3^2 \end{aligned}$$

At fixed material coordinates, the material rate is

$$\begin{aligned} \frac{Df}{Dt} &= \frac{\partial f}{\partial t} = \frac{\partial}{\partial t} [(X_1 + t)^2 + (X_2 + 2t)^2 + X_3^2] \\ &= 2(X_1 + t) + 4(X_2 + 2t) \\ &= 2X_1 + 4X_2 + 10t \end{aligned}$$

To express this same rate in spatial coordinates, invert the motion:

$$X_1 = x_1 - t, \quad X_2 = x_2 - 2t$$

Substitution gives

$$\begin{aligned} \frac{Df}{Dt} &= 2X_1 + 4X_2 + 10t \\ &= 2(x_1 - t) + 4(x_2 - 2t) + 10t \\ &= 2x_1 + 4x_2. \end{aligned}$$

The two final expressions describe the same material rate; only their independent coordinates differ.

2.3 Material derivative

The time derivative of a quantity f experienced by a material particle is called the **material derivative**. When f is written in material coordinates,

$$f = f(X_1, X_2, X_3, t) \tag{2.4}$$

and the particle label $\mathbf{X}$ is held fixed:

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} \tag{2.5}$$

When the spatial description is used instead,

$$f = f(x_1, x_2, x_3, t) \tag{2.6}$$

where the spatial coordinates are related to the material coordinates through the motion

$$x_1 = x_1(X_1, X_2, X_3, t), \quad x_2 = x_2(X_1, X_2, X_3, t), \quad x_3 = x_3(X_1, X_2, X_3, t) \quad (2.7)$$

the chain rule gives

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t} + \frac{\partial f}{\partial x_3} \frac{\partial x_3}{\partial t} \quad (2.8)$$

At fixed $\mathbf{X}$, the derivatives $\frac{\partial x_1}{\partial t}$, $\frac{\partial x_2}{\partial t}$, and $\frac{\partial x_3}{\partial t}$ are the components of the particle velocity $\mathbf{v}$. In Cartesian coordinates, the material derivative becomes

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + v_1 \frac{\partial f}{\partial x_1} + v_2 \frac{\partial f}{\partial x_2} + v_3 \frac{\partial f}{\partial x_3} \quad (2.9)$$

or, in direct notation,

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f \quad (2.10)$$

Here, $\frac{\partial f}{\partial t}$ is evaluated at fixed $\mathbf{x}$ and accounts for the local change of the field. The term $\mathbf{v} \cdot \nabla f$ is the **convective contribution**, caused by the particle moving through a spatially varying field.

This form is valid for fields expressed in spatial coordinates. More generally, the expression remains valid in any coordinate system if the appropriate gradient operator is used.

The material derivative therefore connects the Lagrangian and Eulerian descriptions: it measures the rate of change experienced by a moving particle in a field that may also vary explicitly with time.

Example 2.3: Find the material derivative of the spatial temperature field $T = \alpha(x_1 + x_2)$ for the motion

$$x_1 = X_1 + ktX_2, \quad x_2 = (1 + kt)X_2, \quad x_3 = X_3$$

Solution:

At fixed $\mathbf{X}$, the velocity components are

$$v_1 = \frac{\partial x_1}{\partial t} = kX_2, \quad v_2 = \frac{\partial x_2}{\partial t} = kX_2, \quad v_3 = \frac{\partial x_3}{\partial t} = 0$$

The spatial gradient of temperature has components

$$\frac{\partial T}{\partial x_1} = \alpha, \quad \frac{\partial T}{\partial x_2} = \alpha, \quad \frac{\partial T}{\partial x_3} = 0$$

Moreover, $\frac{\partial T}{\partial t} = 0$ at fixed $\mathbf{x}$ because the spatial expression for T has no explicit time dependence. Therefore,

$$\begin{aligned} \frac{DT}{Dt} &= \frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \\ &= \frac{\partial T}{\partial t} + v_1 \frac{\partial T}{\partial x_1} + v_2 \frac{\partial T}{\partial x_2} + v_3 \frac{\partial T}{\partial x_3} \\ &= kX_2\alpha + kX_2\alpha \end{aligned}$$

This intermediate result is in material coordinates. Since $X_2 = \frac{x_2}{1+kt}$, its spatial description is

$$\frac{DT}{Dt} = \frac{2k\alpha x_2}{1 + kt}.$$

Thus, the temperature experienced by a particle changes even though the local time derivative at a fixed spatial point is zero.

2.4 Deformation gradient

At a fixed time, the spatial position of the particle labeled by $\mathbf{X} + d\mathbf{X}$ can be expanded about $\mathbf{X}$:

$$x_i(\mathbf{X} + d\mathbf{X}, t) = x_{i(\mathbf{X}, t)} + \frac{\partial x_i}{\partial X_j} dX_j + \frac{1}{2} \frac{\partial^2 x_i}{\partial X_j \partial X_k} dX_j dX_k + \dots \quad (2.11)$$

Retaining only first-order terms gives

$$x_i(\mathbf{X} + d\mathbf{X}, t) - x_{i(\mathbf{X}, t)} = \frac{\partial x_i}{\partial X_j} dX_j. \quad (2.12)$$

In direct notation, the corresponding infinitesimal line elements satisfy

$$d\mathbf{x} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} d\mathbf{X}. \quad (2.13)$$

The second-order tensor

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}}, \quad F_{ij} = \frac{\partial x_i}{\partial X_j} \quad (2.14)$$

is the **deformation gradient**. It locally maps a material line element $d\mathbf{X}$ in the reference configuration to the corresponding spatial line element $d\mathbf{x}$ in the current configuration, as illustrated in [Figure 2.2](#):

$$d\mathbf{x} = \mathbf{F} d\mathbf{X} \quad (2.15)$$

For an orientation-preserving deformation, $\mathbf{F}$ can be uniquely decomposed into a rotation tensor and a symmetric positive-definite stretch tensor. The two forms of the polar decomposition are

$$\mathbf{F} = \mathbf{R}\mathbf{U} \quad \text{Right polar decomposition} \quad (2.16)$$

and

$$\mathbf{F} = \mathbf{V}\mathbf{R} \quad \text{Left polar decomposition} \quad (2.17)$$

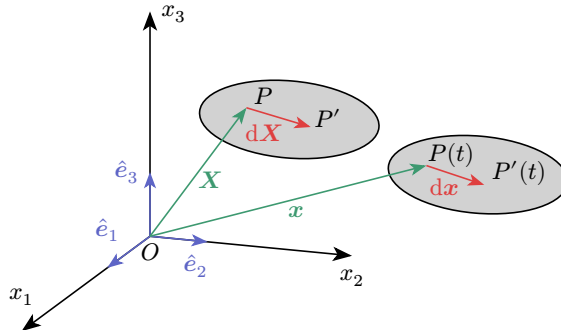


Figure 2.2: Corresponding infinitesimal line elements in the reference and current configurations.

where

- $\mathbf{R}$ is a proper orthogonal tensor that represents the local rigid-body rotation;
- $\mathbf{U}$ is the right stretch tensor, which represents stretching with respect to the reference configuration and
- $\mathbf{V}$ is the left stretch tensor, which represents stretching with respect to the current configuration.

Tensors $\mathbf{U}$ and $\mathbf{V}$ can be calculated as

$$\begin{aligned}\mathbf{U} &= \sqrt{\mathbf{F}^T \mathbf{F}} = \sqrt{\mathbf{C}} \\ \mathbf{V} &= \sqrt{\mathbf{F} \mathbf{F}^T} = \sqrt{\mathbf{B}}\end{aligned}\tag{2.18}$$

The products $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ and $\mathbf{B} = \mathbf{F} \mathbf{F}^T$ are the right and left Cauchy–Green deformation tensors, respectively.

Once $\mathbf{U}$ or $\mathbf{V}$ is known, $\mathbf{R}$ follows from the corresponding polar decomposition. For a rigid-body motion, $\mathbf{U} = \mathbf{V} = \mathbf{I}$ and $\mathbf{R} = \mathbf{F}$. For a pure translation, $\mathbf{F} = \mathbf{I}$.

2.5 Displacement gradient

The displacement field is $\mathbf{u}(\mathbf{X}, t) = \mathbf{x}(\mathbf{X}, t) - \mathbf{X}$, as shown in **Figure 2.3**. Therefore, $\mathbf{F}$ can be written as

$$\begin{aligned}\mathbf{F} &= \frac{\partial \mathbf{x}}{\partial \mathbf{X}} \\ &= \frac{\partial}{\partial \mathbf{X}}(\mathbf{u} + \mathbf{X}) \\ &= \nabla_{\mathbf{X}} \mathbf{u} + \mathbf{I}\end{aligned}\tag{2.19}$$

where $\nabla_{\mathbf{X}}$ denotes the gradient with respect to the material coordinates. The tensor $\nabla_{\mathbf{X}} \mathbf{u}$ is the **displacement gradient** in the reference description.

For a pure translation, $\mathbf{u}$ is constant, so $\nabla_{\mathbf{X}} \mathbf{u} = \mathbf{0}$ and $\mathbf{F} = \mathbf{I}$.

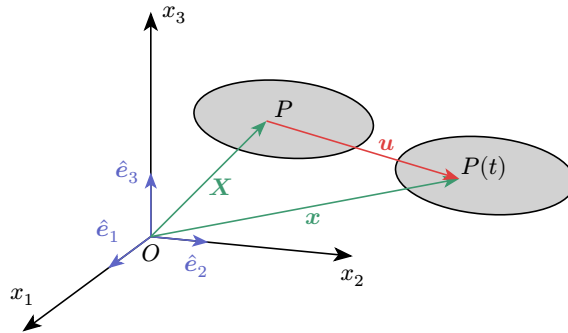


Figure 2.3: Displacement vector between corresponding material points.

2.6 Material strain tensor

To quantify changes in the internal geometry of a body, compare the squared lengths of corresponding line elements in the reference and current configurations. Using $d\mathbf{x} = \mathbf{F} d\mathbf{X}$,

$$\begin{aligned} d\mathbf{x} \cdot d\mathbf{x} - d\mathbf{X} \cdot d\mathbf{X} &= (\mathbf{F} d\mathbf{X}) \cdot (\mathbf{F} d\mathbf{X}) - d\mathbf{X} \cdot d\mathbf{X} \\ &= d\mathbf{X} \cdot (\mathbf{F}^T \mathbf{F} d\mathbf{X}) - d\mathbf{X} \cdot d\mathbf{X} \\ &= d\mathbf{X} \cdot \underbrace{(\mathbf{F}^T \mathbf{F} - \mathbf{I})}_{2\mathbf{E}} d\mathbf{X} \end{aligned} \quad (2.20)$$

This relation defines the **material strain tensor**, also known as the Green–Lagrange strain tensor:

$$\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I}) \quad (2.21)$$

Because $\mathbf{E}$ is expressed in the reference configuration, it measures changes relative to material line elements. Its diagonal components quantify squared-length changes along the selected coordinate directions. These directions are principal directions only when the chosen basis diagonalizes $\mathbf{E}$. The off-diagonal components quantify changes in the inner products between pairs of material directions; for finite deformation, they should not be identified directly with angle changes without accounting for the accompanying stretches.

In terms of the displacement gradient, the Green–Lagrange tensor is

$$\begin{aligned} \mathbf{E} &= \frac{1}{2}((\nabla_{\mathbf{X}} \mathbf{u} + \mathbf{I})^T (\nabla_{\mathbf{X}} \mathbf{u} + \mathbf{I}) - \mathbf{I}) \\ &= \frac{1}{2}(\nabla_{\mathbf{X}} \mathbf{u} + \nabla_{\mathbf{X}} \mathbf{u}^T + \nabla_{\mathbf{X}} \mathbf{u}^T \nabla_{\mathbf{X}} \mathbf{u}). \end{aligned} \quad (2.22)$$

2.7 Spatial strain tensor

An Eulerian strain measure is obtained by expressing the same change in squared length using the current line element. Since

$$d\mathbf{X} = \mathbf{F}^{-1} d\mathbf{x} \quad (2.23)$$

the difference is

$$\begin{aligned} d\mathbf{x} \cdot d\mathbf{x} - d\mathbf{X} \cdot d\mathbf{X} &= d\mathbf{x} \cdot d\mathbf{x} - (\mathbf{F}^{-1} d\mathbf{x}) \cdot (\mathbf{F}^{-1} d\mathbf{x}) \\ &= d\mathbf{x} \cdot d\mathbf{x} - d\mathbf{x} \cdot (\mathbf{F}^{-T} \mathbf{F}^{-1} d\mathbf{x}) \\ &= d\mathbf{x} \cdot \underbrace{[\mathbf{I} - (\mathbf{F}^{-T} \mathbf{F}^{-1})]}_{2\mathbf{e}} d\mathbf{x} \end{aligned} \quad (2.24)$$

where $\mathbf{e}$ is the **spatial strain tensor**, also known as the Eulerian–Almansi strain tensor:

$$\mathbf{e} = \frac{1}{2}(\mathbf{I} - (\mathbf{F} \mathbf{F}^T)^{-1}) \quad (2.25)$$

The Almansi tensor describes deformation in the current configuration and is useful in Eulerian formulations of finite deformation. It is distinct from logarithmic strain, which is sometimes called **true strain**.

2.8 Infinitesimal deformation

The Green–Lagrange strain tensor was defined in terms of the displacement field $\mathbf{u}$ as

$$\mathbf{E} = \frac{1}{2}(\nabla_{\mathbf{X}} \mathbf{u} + \nabla_{\mathbf{X}} \mathbf{u}^T + \nabla_{\mathbf{X}} \mathbf{u}^T \nabla_{\mathbf{X}} \mathbf{u}) \quad (2.26)$$

If every component of the displacement gradient is small compared with unity, the quadratic term is of higher order and may be neglected. The Green–Lagrange tensor then reduces to the **infinitesimal strain tensor**

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla_{\mathbf{X}} \mathbf{u} + \nabla_{\mathbf{X}} \mathbf{u}^T) \quad (2.27)$$

This tensor is the symmetric part of the displacement gradient. The components of $\boldsymbol{\varepsilon}$ are

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right) \quad (2.28)$$

The tensor $\boldsymbol{\varepsilon}$ can be written in full form as

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \frac{\partial u_1}{\partial X_1} & \frac{1}{2} \left(\frac{\partial u_1}{\partial X_2} + \frac{\partial u_2}{\partial X_1} \right) & \frac{1}{2} \left(\frac{\partial u_1}{\partial X_3} + \frac{\partial u_3}{\partial X_1} \right) \\ \frac{1}{2} \left(\frac{\partial u_2}{\partial X_1} + \frac{\partial u_1}{\partial X_2} \right) & \frac{\partial u_2}{\partial X_2} & \frac{1}{2} \left(\frac{\partial u_2}{\partial X_3} + \frac{\partial u_3}{\partial X_2} \right) \\ \frac{1}{2} \left(\frac{\partial u_3}{\partial X_1} + \frac{\partial u_1}{\partial X_3} \right) & \frac{1}{2} \left(\frac{\partial u_3}{\partial X_2} + \frac{\partial u_2}{\partial X_3} \right) & \frac{\partial u_3}{\partial X_3} \end{pmatrix} \quad (2.29)$$

Figure 2.4 shows a representation of the components of the strain tensor $\boldsymbol{\varepsilon}$.

Note 2.1: When the displacement gradient is small, differences between the reference and current coordinates are of higher order, and $\boldsymbol{\varepsilon} \approx \mathbf{E} \approx \mathbf{e}$. Small strain alone is not sufficient for this linearization if finite rotations are present.

The antisymmetric part of the displacement gradient is the **infinitesimal rotation tensor**:

$$\boldsymbol{\omega} = \frac{1}{2}(\nabla_{\mathbf{X}} \mathbf{u} - \nabla_{\mathbf{X}} \mathbf{u}^T) \quad (2.30)$$

which can be written in full form as

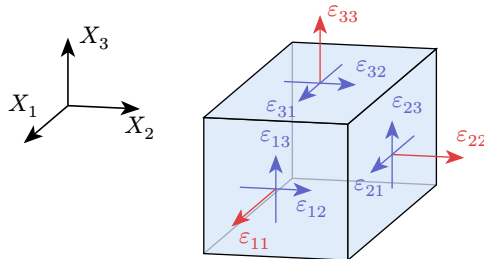
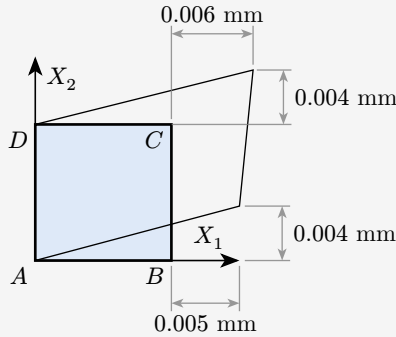


Figure 2.4: Components of the infinitesimal strain tensor.

$$\boldsymbol{\omega} = \begin{pmatrix} 0 & \frac{1}{2}\left(\frac{\partial u_1}{\partial X_2} - \frac{\partial u_2}{\partial X_1}\right) & \frac{1}{2}\left(\frac{\partial u_1}{\partial X_3} - \frac{\partial u_3}{\partial X_1}\right) \\ \frac{1}{2}\left(\frac{\partial u_2}{\partial X_1} - \frac{\partial u_1}{\partial X_2}\right) & 0 & \frac{1}{2}\left(\frac{\partial u_2}{\partial X_3} - \frac{\partial u_3}{\partial X_2}\right) \\ \frac{1}{2}\left(\frac{\partial u_3}{\partial X_1} - \frac{\partial u_1}{\partial X_3}\right) & \frac{1}{2}\left(\frac{\partial u_3}{\partial X_2} - \frac{\partial u_2}{\partial X_3}\right) & 0 \end{pmatrix} \quad (2.31)$$

Tensor $\boldsymbol{\omega}$ describes the local rotation to first order. Unlike $\boldsymbol{\varepsilon}$, it does not change lengths or angles to first order and therefore does not contribute to strain energy in infinitesimal elasticity.

Example 2.4: The figure shows a square body with side length 1 mm in the reference configuration and prescribed corner displacements. Assuming a bilinear interpolation, determine the displacement field $\mathbf{u}$. Then evaluate the Green–Lagrange strain tensor $\mathbf{E}$, the infinitesimal strain tensor $\boldsymbol{\varepsilon}$, and the infinitesimal rotation tensor $\boldsymbol{\omega}$ at the material point $\mathbf{X} = (1, 1, 0)$ mm. Assume that $u_3 = 0$.



Solution:

Using millimetres for both the material coordinates and the displacements, assume

$$\begin{aligned} u_1(X_1, X_2) &= c_1 + c_2 X_1 + c_3 X_2 + c_4 X_1 X_2 \\ u_2(X_1, X_2) &= c_5 + c_6 X_1 + c_7 X_2 + c_8 X_1 X_2 \end{aligned}$$

The prescribed horizontal displacements at A , B , C , and D give

$$\begin{aligned} u_1(A) &= c_1 = 0 \\ u_1(B) &= c_1 + c_2 = 0.005 \\ u_1(C) &= c_1 + c_2 + c_3 + c_4 = 0.006 \\ u_1(D) &= c_1 + c_3 = 0 \end{aligned}$$

Hence,

$$c_1 = 0, c_2 = 0.005, c_3 = 0, c_4 = 0.001$$

Similarly, the prescribed vertical displacements give

$$\begin{aligned} u_2(A) &= c_5 = 0 \\ u_2(B) &= c_5 + c_6 = 0.004 \\ u_2(C) &= c_5 + c_6 + c_7 + c_8 = 0.004 \\ u_2(D) &= c_5 + c_7 = 0 \end{aligned}$$

and therefore

$$c_5 = 0, c_6 = 0.004, c_7 = 0, c_8 = 0$$

The displacement field is thus

$$\mathbf{u} = (0.005X_1 + 0.001X_1X_2)\hat{\mathbf{e}}_1 + 0.004X_1\hat{\mathbf{e}}_2.$$

Here, the coefficients carry the units required for $\mathbf{u}$ to be measured in millimetres; the resulting displacement gradients are dimensionless.

The displacement gradient is

$$\nabla_{\mathbf{X}} \mathbf{u} = \begin{pmatrix} 0.005 + 0.001X_2 & 0.001X_1 & 0 \\ 0.004 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

At $\mathbf{X} = (1, 1, 0)$ mm,

$$\nabla_{\mathbf{X}} \mathbf{u} = \begin{pmatrix} 0.006 & 0.001 & 0 \\ 0.004 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Using the quadratic product $\nabla_{\mathbf{X}} \mathbf{u}^T \nabla_{\mathbf{X}} \mathbf{u}$, the Green–Lagrange strain tensor is

$$\mathbf{E} = \begin{pmatrix} 0.006026 & 0.002503 & 0 \\ 0.002503 & 0.0000005 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Retaining only the symmetric linear terms gives

$$\boldsymbol{\varepsilon} = \begin{pmatrix} 0.006 & 0.0025 & 0 \\ 0.0025 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The antisymmetric part gives the infinitesimal rotation tensor

$$\boldsymbol{\omega} = \begin{pmatrix} 0 & -0.0015 & 0 \\ 0.0015 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

As a check, $\nabla_{\mathbf{X}} \mathbf{u} = \boldsymbol{\varepsilon} + \boldsymbol{\omega}$, while the small difference between $\mathbf{E}$ and $\boldsymbol{\varepsilon}$ is the retained quadratic contribution.

Exercises 2.1:

1. A body is subjected to the displacement field

$$\mathbf{u} = (X_1^2 + X_2^2)\hat{\mathbf{e}}_1 + X_1X_2\hat{\mathbf{e}}_2$$

Determine the Green–Lagrange strain tensor $\mathbf{E}$, the infinitesimal strain tensor $\boldsymbol{\varepsilon}$, and the infinitesimal rotation tensor $\boldsymbol{\omega}$.

2.9 Compatibility equations

For a sufficiently smooth displacement field $\mathbf{u}$, the infinitesimal strain field follows by differentiation:

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right) \quad (2.32)$$

The converse is not automatic: an arbitrary symmetric tensor field need not be the symmetric gradient of a continuous, single-valued displacement field. A strain field is called **compatible** when such a displacement field exists, so that the deformation introduces no gaps or overlaps within the body. Compatibility is first illustrated by the analogous problem of recovering a scalar potential from a vector field.

Example 2.5: A vector field $\mathbf{v}$ is a potential field if there is a scalar field φ such that

$$\mathbf{v} = \nabla \varphi$$

Find the compatibility conditions that $\mathbf{v}$ must satisfy.

Solution:

In components,

$$v_1 = \frac{\partial \varphi}{\partial X_1}, \quad v_2 = \frac{\partial \varphi}{\partial X_2}, \quad v_3 = \frac{\partial \varphi}{\partial X_3}.$$

Differentiate the first and second equations with respect to X_2 and X_1 , respectively; repeat the same procedure for the pairs (1, 3) and (2, 3):

$$\begin{aligned} \frac{\partial v_1}{\partial X_2} &= \frac{\partial^2 \varphi}{\partial X_2 \partial X_1}, & \frac{\partial v_2}{\partial X_1} &= \frac{\partial^2 \varphi}{\partial X_1 \partial X_2} \\ \frac{\partial v_1}{\partial X_3} &= \frac{\partial^2 \varphi}{\partial X_3 \partial X_1}, & \frac{\partial v_3}{\partial X_1} &= \frac{\partial^2 \varphi}{\partial X_1 \partial X_3} \\ \frac{\partial v_2}{\partial X_3} &= \frac{\partial^2 \varphi}{\partial X_3 \partial X_2}, & \frac{\partial v_3}{\partial X_2} &= \frac{\partial^2 \varphi}{\partial X_2 \partial X_3}. \end{aligned}$$

If φ has continuous second derivatives, Schwarz's theorem gives

$$\frac{\partial^2 \varphi}{\partial X_2 \partial X_1} = \frac{\partial^2 \varphi}{\partial X_1 \partial X_2}, \quad \frac{\partial^2 \varphi}{\partial X_3 \partial X_1} = \frac{\partial^2 \varphi}{\partial X_1 \partial X_3}, \quad \frac{\partial^2 \varphi}{\partial X_3 \partial X_2} = \frac{\partial^2 \varphi}{\partial X_2 \partial X_3}.$$

Subtracting each pair of differentiated equations therefore gives

$$\frac{\partial v_1}{\partial X_2} - \frac{\partial v_2}{\partial X_1} = 0, \quad \frac{\partial v_1}{\partial X_3} - \frac{\partial v_3}{\partial X_1} = 0, \quad \frac{\partial v_2}{\partial X_3} - \frac{\partial v_3}{\partial X_2} = 0$$

or, equivalently, $\nabla \times \mathbf{v} = \mathbf{0}$. These conditions are necessary; they are also sufficient when $\mathbf{v}$ is continuously differentiable on a simply connected domain.

Returning to the infinitesimal strain tensor,

$$\varepsilon = \begin{pmatrix} \frac{\partial u_1}{\partial X_1} & \frac{1}{2} \left(\frac{\partial u_1}{\partial X_2} + \frac{\partial u_2}{\partial X_1} \right) & \frac{1}{2} \left(\frac{\partial u_1}{\partial X_3} + \frac{\partial u_3}{\partial X_1} \right) \\ \frac{1}{2} \left(\frac{\partial u_2}{\partial X_1} + \frac{\partial u_1}{\partial X_2} \right) & \frac{\partial u_2}{\partial X_2} & \frac{1}{2} \left(\frac{\partial u_2}{\partial X_3} + \frac{\partial u_3}{\partial X_2} \right) \\ \frac{1}{2} \left(\frac{\partial u_3}{\partial X_1} + \frac{\partial u_1}{\partial X_3} \right) & \frac{1}{2} \left(\frac{\partial u_3}{\partial X_2} + \frac{\partial u_2}{\partial X_3} \right) & \frac{\partial u_3}{\partial X_3} \end{pmatrix} \quad (2.33)$$

only six components are independent because ε is symmetric. The strain–displacement equations are

$$\begin{aligned} \varepsilon_{11} - \frac{\partial u_1}{\partial X_1} &= 0, & \varepsilon_{12} - \frac{1}{2} \left(\frac{\partial u_1}{\partial X_2} + \frac{\partial u_2}{\partial X_1} \right) &= 0, \\ \varepsilon_{22} - \frac{\partial u_2}{\partial X_2} &= 0, & \varepsilon_{13} - \frac{1}{2} \left(\frac{\partial u_1}{\partial X_3} + \frac{\partial u_3}{\partial X_1} \right) &= 0, \\ \varepsilon_{33} - \frac{\partial u_3}{\partial X_3} &= 0, & \varepsilon_{23} - \frac{1}{2} \left(\frac{\partial u_2}{\partial X_3} + \frac{\partial u_3}{\partial X_2} \right) &= 0 \end{aligned} \quad (2.34)$$

Eliminating the three displacement components by differentiating these relations and using equality of mixed derivatives gives the six Saint-Venant compatibility equations:

$$\begin{aligned} \frac{\partial^2 \varepsilon_{11}}{\partial X_2^2} + \frac{\partial^2 \varepsilon_{22}}{\partial X_1^2} - 2 \frac{\partial^2 \varepsilon_{12}}{\partial X_1 \partial X_2} &= 0 \\ \frac{\partial^2 \varepsilon_{11}}{\partial X_3^2} + \frac{\partial^2 \varepsilon_{33}}{\partial X_1^2} - 2 \frac{\partial^2 \varepsilon_{13}}{\partial X_1 \partial X_3} &= 0 \\ \frac{\partial^2 \varepsilon_{22}}{\partial X_3^2} + \frac{\partial^2 \varepsilon_{33}}{\partial X_2^2} - 2 \frac{\partial^2 \varepsilon_{23}}{\partial X_2 \partial X_3} &= 0 \\ \frac{\partial^2 \varepsilon_{11}}{\partial X_2 \partial X_3} + \frac{\partial^2 \varepsilon_{23}}{\partial X_1^2} - \frac{\partial^2 \varepsilon_{13}}{\partial X_1 \partial X_2} - \frac{\partial^2 \varepsilon_{12}}{\partial X_1 \partial X_3} &= 0 \\ \frac{\partial^2 \varepsilon_{22}}{\partial X_1 \partial X_3} + \frac{\partial^2 \varepsilon_{13}}{\partial X_2^2} - \frac{\partial^2 \varepsilon_{12}}{\partial X_2 \partial X_3} - \frac{\partial^2 \varepsilon_{23}}{\partial X_1 \partial X_2} &= 0 \\ \frac{\partial^2 \varepsilon_{33}}{\partial X_1 \partial X_2} + \frac{\partial^2 \varepsilon_{12}}{\partial X_3^2} - \frac{\partial^2 \varepsilon_{23}}{\partial X_1 \partial X_3} - \frac{\partial^2 \varepsilon_{13}}{\partial X_2 \partial X_3} &= 0. \end{aligned} \quad (2.35)$$

For a twice continuously differentiable strain field on a simply connected domain, these equations are sufficient for the existence of a displacement field, up to an arbitrary infinitesimal rigid-body motion.

For a two-dimensional strain field involving ε_{11} , ε_{22} , and ε_{12} , the equations reduce to

$$\frac{\partial^2 \varepsilon_{11}}{\partial X_2^2} + \frac{\partial^2 \varepsilon_{22}}{\partial X_1^2} - 2 \frac{\partial^2 \varepsilon_{12}}{\partial X_1 \partial X_2} = 0 \quad (2.36)$$

Example 2.6: Given the two-dimensional infinitesimal strain field, with $k \neq 0$,

$$\varepsilon = \begin{pmatrix} 0 & kX_1X_2 & 0 \\ kX_1X_2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

determine whether the compatibility equation is satisfied. Then show directly that no sufficiently smooth displacement field corresponds to this strain field.

Solution:

The two-dimensional compatibility residual is

$$C = \frac{\partial^2 \varepsilon_{11}}{\partial X_2^2} + \frac{\partial^2 \varepsilon_{22}}{\partial X_1^2} - 2 \frac{\partial^2 \varepsilon_{12}}{\partial X_1 \partial X_2}$$

Compatibility requires $C = 0$, but

$$\begin{aligned} C &= \frac{\partial^2 0}{\partial X_2^2} + \frac{\partial^2 0}{\partial X_1^2} - 2 \frac{\partial^2}{\partial X_1 \partial X_2} (kX_1 X_2) \\ &= -2k \neq 0. \end{aligned}$$

The same incompatibility can be seen by attempting to recover the displacement. From

$$\varepsilon_{11} = \frac{\partial u_1}{\partial X_1} = 0, \quad \varepsilon_{22} = \frac{\partial u_2}{\partial X_2} = 0,$$

integration gives

$$u_1 = f(X_2), \quad u_2 = g(X_1).$$

The shear component would then require

$$\frac{\partial f}{\partial X_2} + \frac{\partial g}{\partial X_1} = 2kX_1 X_2.$$

Differentiating this equation with respect to both X_1 and X_2 produces

$$0 = 2k,$$

which is impossible for $k \neq 0$. Therefore, no such displacement field exists.

Exercises 2.2:

1. Determine whether the following tensor field is a compatible infinitesimal strain field:

$$\varepsilon = \begin{pmatrix} kX_2^2 & kX_1 & 0 \\ kX_1 & kX_2 & 0 \\ 0 & 0 & kX_2^2 \end{pmatrix}$$

2. Show that the tensor field below is compatible:

$$\varepsilon = \begin{pmatrix} 2kX_1 & \frac{3}{2}kX_2 & 0 \\ \frac{3}{2}kX_2 & kX_1 & 0 \\ 0 & 0 & kX_3 \end{pmatrix}$$

2.10 Typical strain states

Several idealized strain states occur frequently in solid mechanics:

Uniaxial strain: Only one normal strain component is nonzero. For example,

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (2.37)$$

Pure shear strain: In a suitable basis, only a pair of symmetric shear components is nonzero:

$$\boldsymbol{\varepsilon} = \begin{pmatrix} 0 & \varepsilon_{12} & 0 \\ \varepsilon_{21} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{or} \quad \boldsymbol{\varepsilon}' = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & -\varepsilon & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (2.38)$$

where $\varepsilon_{12} = \varepsilon_{21}$ and $\varepsilon = \varepsilon'_{11} = -\varepsilon'_{22}$. The two tensors represent the same strain state in coordinate systems rotated by 45° . A simple-shear deformation produces this infinitesimal strain together with an infinitesimal rotation, so **simple shear** and **pure shear** are not synonymous descriptions of the motion.

Plane strain: All strain components involving the out-of-plane direction vanish. For deformation in the X_1 – X_2 plane,

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} & 0 \\ \varepsilon_{21} & \varepsilon_{22} & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (2.39)$$

Spherical strain: Equal normal strains act in all three directions, with no shear components:

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{pmatrix} \quad (2.40)$$

where $\varepsilon = \varepsilon_{11} = \varepsilon_{22} = \varepsilon_{33}$.

2.11 Volumetric strain

Volumetric strain, also called **dilatation**, is the relative change in the volume of a material element. For finite deformation, the exact local volume ratio is $J = \det(\mathbf{F})$, so

$$\frac{\Delta(dV)}{dV} = J - 1, \quad (2.41)$$

where dV is an infinitesimal volume element in the reference configuration. For infinitesimal kinematics, $\mathbf{F} = \mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u}$ and the determinant can be linearized as

$$J = \det(\mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u}) \approx 1 + \text{trace}(\nabla_{\mathbf{X}} \mathbf{u}). \quad (2.42)$$

Because the antisymmetric part of the displacement gradient has zero trace,

$$\varepsilon_v = \frac{\Delta(dV)}{dV} \approx \text{trace}(\boldsymbol{\varepsilon}) = \varepsilon_{ii} = \frac{\partial u_i}{\partial X_i} = \nabla \cdot \mathbf{u}. \quad (2.43)$$

Equivalently, $\varepsilon_v = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$ in the principal basis. Since the trace is invariant, the same value is obtained in every Cartesian basis.

Thus, the first invariant $I_1 = \text{trace}(\varepsilon)$ determines whether the material element expands ($I_1 > 0$) or contracts ($I_1 < 0$) to first order. A volume change need not be uniform in all directions.

Exercises 2.3:

1. Given the displacement field

$$\mathbf{u} = k(X_1^2 + X_2^2)t \hat{\mathbf{e}}_1 + kX_1X_2\hat{\mathbf{e}}_2 + kX_1X_3\hat{\mathbf{e}}_3$$

where $k = 10^{-4}$, determine the volumetric strain at point $\mathbf{X} = (2, 1, 1)$ for $t = 2$.

2.12 Engineering strains

Engineering strains describe relative changes in length and, within infinitesimal kinematics, changes in angle between initially perpendicular material lines. The normal engineering strain in the x direction is

$$\varepsilon_x = \frac{\Delta L_x}{L_x} = \varepsilon_{11} \quad (2.44)$$

where L_x and ΔL_x are the original length and its change, respectively.

The engineering shear strain in the xy plane is

$$\gamma_{xy} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} = 2\varepsilon_{12}. \quad (2.45)$$

With the usual sign convention, γ_{xy} is the decrease in the right angle between those lines, measured in radians. Relations such as $\tan(\gamma_{xy}) \approx \gamma_{xy}$ are small-angle approximations, not the definition of finite shear strain. The factor of two follows from symmetrizing the displacement gradient.

The engineering components are therefore related to the infinitesimal strain tensor by

$$\varepsilon = \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{pmatrix} = \begin{pmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz} \\ \frac{1}{2}\gamma_{yx} & \varepsilon_y & \frac{1}{2}\gamma_{yz} \\ \frac{1}{2}\gamma_{zx} & \frac{1}{2}\gamma_{zy} & \varepsilon_z \end{pmatrix} \quad (2.46)$$

Because the strain tensor is symmetric, its six independent engineering components are commonly arranged in the vector

$$\boldsymbol{\varepsilon}_{\text{eng}} = \begin{pmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{yz} \\ \gamma_{zx} \\ \gamma_{xy} \end{pmatrix} \quad (2.47)$$

Note 2.2: The ordering of the shear components varies among sources. All transformation and constitutive matrices must use the same ordering as the associated strain vector.

2.13 Strain transformation

Let $\hat{e}'_1$, $\hat{e}'_2$, and $\hat{e}'_3$ be the new orthonormal basis vectors expressed in the original basis. Collecting them as the columns of $\mathbf{R}$ gives

$$R_{ij} = \hat{e}_i \cdot \hat{e}'_j, \quad \mathbf{R} = (\hat{e}'_1 \mid \hat{e}'_2 \mid \hat{e}'_3). \quad (2.48)$$

The strain components in the new basis are then

$$\boldsymbol{\varepsilon}' = \mathbf{R}^T \boldsymbol{\varepsilon} \mathbf{R}. \quad (2.49)$$

Thus,

$$\begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix} = \begin{pmatrix} \mid & \mid & \mid \\ \hat{e}'_1 & \hat{e}'_2 & \hat{e}'_3 \\ \mid & \mid & \mid \end{pmatrix} \quad (2.50)$$

The columns—not the rows—are the new basis vectors. Equivalently, the entries of each column are the direction cosines of one new basis vector relative to the original basis.

For the engineering strain-vector ordering in Eq. (2.47), the same transformation can be written

$$\boldsymbol{\varepsilon}'_{\text{eng}} = \bar{\mathbf{R}} \boldsymbol{\varepsilon}_{\text{eng}}, \quad (2.51)$$

where $\bar{\mathbf{R}}$ is a 6×6 matrix constructed from the direction cosines. Define

$$\mathbf{R} = \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix} = \begin{pmatrix} \mid & \mid & \mid \\ \mathbf{l} & \mathbf{m} & \mathbf{n} \\ \mid & \mid & \mid \end{pmatrix} \quad (2.52)$$

where $\mathbf{l} = \hat{e}'_1$, $\mathbf{m} = \hat{e}'_2$, and $\mathbf{n} = \hat{e}'_3$, all expressed in the original basis. The engineering-strain transformation matrix is

$$\bar{\mathbf{R}} = \begin{pmatrix} l_1^2 & l_2^2 & l_3^2 & l_2 l_3 & l_3 l_1 & l_1 l_2 \\ m_1^2 & m_2^2 & m_3^2 & m_2 m_3 & m_3 m_1 & m_1 m_2 \\ n_1^2 & n_2^2 & n_3^2 & n_2 n_3 & n_3 n_1 & n_1 n_2 \\ 2m_1 n_1 & 2m_2 n_2 & 2m_3 n_3 & m_2 n_3 + m_3 n_2 & m_3 n_1 + m_1 n_3 & m_1 n_2 + m_2 n_1 \\ 2l_1 n_1 & 2l_2 n_2 & 2l_3 n_3 & l_3 n_2 + n_3 l_2 & l_1 n_3 + n_1 l_3 & l_2 n_1 + n_2 l_1 \\ 2l_1 m_1 & 2l_2 m_2 & 2l_3 m_3 & l_2 m_3 + l_3 m_2 & l_3 m_1 + l_1 m_3 & l_1 m_2 + l_2 m_1 \end{pmatrix} \quad (2.53)$$

Note 2.3: The matrix $\bar{\mathbf{R}}$ is specific to engineering shear components and to the ordering in Eq. (2.47). It must not be applied to a vector containing the tensorial shear components ε_{ij} .

In Mandel notation, define

$$\boldsymbol{\varepsilon}_M = \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \sqrt{2}\varepsilon_{23} \\ \sqrt{2}\varepsilon_{31} \\ \sqrt{2}\varepsilon_{12} \end{pmatrix}. \quad (2.54)$$

The corresponding transformation matrix is

$$\bar{\mathbf{R}}_M = \begin{pmatrix} l_1^2 & l_2^2 & l_3^2 & \sqrt{2}l_2l_3 & \sqrt{2}l_3l_1 & \sqrt{2}l_1l_2 \\ m_1^2 & m_2^2 & m_3^2 & \sqrt{2}m_2m_3 & \sqrt{2}m_3m_1 & \sqrt{2}m_1m_2 \\ n_1^2 & n_2^2 & n_3^2 & \sqrt{2}n_2n_3 & \sqrt{2}n_3n_1 & \sqrt{2}n_1n_2 \\ \sqrt{2}m_1n_1 & \sqrt{2}m_2n_2 & \sqrt{2}m_3n_3 & m_2n_3 + m_3n_2 & m_3n_1 + m_1n_3 & m_1n_2 + m_2n_1 \\ \sqrt{2}l_1n_1 & \sqrt{2}l_2n_2 & \sqrt{2}l_3n_3 & l_3n_2 + n_3l_2 & l_1n_3 + n_1l_3 & l_2n_1 + n_2l_1 \\ \sqrt{2}l_1m_1 & \sqrt{2}l_2m_2 & \sqrt{2}l_3m_3 & l_2m_3 + l_3m_2 & l_3m_1 + l_1m_3 & l_1m_2 + l_2m_1 \end{pmatrix} \quad (2.55)$$

For a two-dimensional coordinate system rotated counterclockwise by θ about the x_3 axis,

$$\begin{aligned} \varepsilon'_{11} &= \frac{\varepsilon_{11} + \varepsilon_{22}}{2} + \frac{\varepsilon_{11} - \varepsilon_{22}}{2} \cos(2\theta) + \varepsilon_{12} \sin(2\theta) \\ \varepsilon'_{22} &= \frac{\varepsilon_{11} + \varepsilon_{22}}{2} - \frac{\varepsilon_{11} - \varepsilon_{22}}{2} \cos(2\theta) - \varepsilon_{12} \sin(2\theta) \\ \varepsilon'_{12} &= -\frac{\varepsilon_{11} - \varepsilon_{22}}{2} \sin(2\theta) + \varepsilon_{12} \cos(2\theta) \end{aligned} \quad (2.56)$$

Example 2.7: Given the following strain tensor, determine its components in a coordinate system rotated by 30° around the x_1 axis in the original coordinate system.

$$\varepsilon = \begin{pmatrix} 0.01 & 0.02 & 0 \\ 0.02 & 0.04 & 0.05 \\ 0 & 0.05 & 0.06 \end{pmatrix}$$

Solution:

The new basis vectors, expressed in the original basis, are

$$\begin{aligned} \hat{e}'_2 &= \cos(30^\circ)\hat{e}_2 + \sin(30^\circ)\hat{e}_3 = 0.866\hat{e}_2 + 0.5\hat{e}_3 \\ \hat{e}'_3 &= -\sin(30^\circ)\hat{e}_2 + \cos(30^\circ)\hat{e}_3 = -0.5\hat{e}_2 + 0.866\hat{e}_3 \end{aligned}$$

Therefore, $\mathbf{R}$ has these vectors as its columns:

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{pmatrix}$$

The strain components in the new coordinate system are

$$\varepsilon' = \mathbf{R}^T \varepsilon \mathbf{R} = \begin{pmatrix} 0.01 & 0.0173 & -0.01 \\ 0.0173 & 0.0883 & 0.0337 \\ -0.01 & 0.0337 & 0.0117 \end{pmatrix}$$

Substituting the same direction cosines into the engineering transformation matrix gives

$$\bar{\mathbf{R}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.75 & 0.25 & 0.433 & 0 & 0 \\ 0 & 0.25 & 0.75 & -0.433 & 0 & 0 \\ 0 & -0.866 & 0.866 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.866 & -0.5 \\ 0 & 0 & 0 & 0 & 0.5 & 0.866 \end{pmatrix}$$

Hence, the engineering strain vector in the new coordinate system is

$$\boldsymbol{\varepsilon}'_{\text{eng}} = \bar{\mathbf{R}} \boldsymbol{\varepsilon}_{\text{eng}} = \begin{pmatrix} 0.01 \\ 0.0883 \\ 0.0117 \\ 0.0673 \\ -0.02 \\ 0.0346 \end{pmatrix}$$

The tensor and engineering-vector calculations agree: each engineering shear component is twice the corresponding off-diagonal tensor component.

2.14 Principal strains and directions

The principal strains are the eigenvalues of the symmetric infinitesimal strain tensor $\boldsymbol{\varepsilon}$. Their associated eigenvectors are mutually orthogonal principal directions along which the shear components vanish, as illustrated in [Figure 2.5](#). They satisfy

$$\boldsymbol{\varepsilon} \mathbf{v}_i = \varepsilon_i \mathbf{v}_i \quad (2.57)$$

where $\mathbf{v}_i$ is the eigenvector associated with ε_i , and no summation is implied on i . The largest and smallest eigenvalues are the maximum and minimum normal strains at the point. Repeated eigenvalues have a multidimensional eigenspace, so their principal directions are not unique.

We order the principal strains as $\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$ and pair each eigenvalue with its corresponding unit eigenvector $\mathbf{v}_i$.

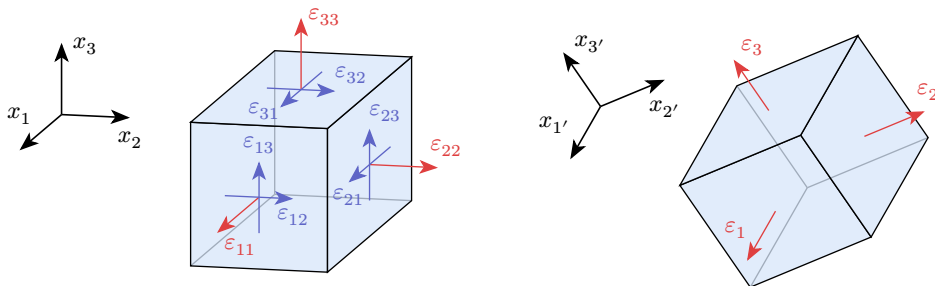
The first invariant of tensor $\boldsymbol{\varepsilon}$ is given by

$$I_1 = \text{trace}(\boldsymbol{\varepsilon}) = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 \quad (2.58)$$

As shown previously, I_1 is the infinitesimal volumetric strain. The other invariants, together with I_1 , characterize the strain tensor independently of the coordinate system.

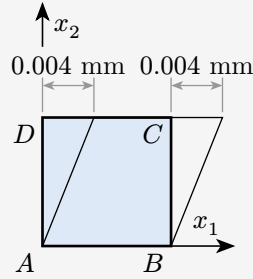
For a two-dimensional strain tensor, the two in-plane principal strains are

$$\varepsilon_{1,2} = \frac{1}{2}(\varepsilon_{11} + \varepsilon_{22}) \pm \sqrt{\left(\frac{\varepsilon_{11} - \varepsilon_{22}}{2}\right)^2 + \varepsilon_{12}^2} \quad (2.59)$$



[Figure 2.5](#): Strain components in the original and principal coordinate systems.

Example 2.8: Find the principal strains and directions for the body under simple shear shown in the figure. The body has side lengths of 1 mm in the x_1 and x_2 directions, and all out-of-plane strain components vanish.



Solution:

The displacement components are

$$u_1 = 0.004X_2, \quad u_2 = 0$$

Then, the strain components are

$$\varepsilon_{11} = \frac{\partial u_1}{\partial X_1} = 0, \quad \varepsilon_{22} = \frac{\partial u_2}{\partial X_2} = 0, \quad \varepsilon_{12} = \frac{1}{2} \left(\frac{\partial u_1}{\partial X_2} + \frac{\partial u_2}{\partial X_1} \right) = 0.002$$

and the strain tensor is given by

$$\varepsilon = \begin{pmatrix} 0 & 0.002 & 0 \\ 0.002 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Ordering all three eigenvalues from largest to smallest gives

$$\varepsilon_1 = 0.002, \quad \varepsilon_2 = 0, \quad \varepsilon_3 = -0.002,$$

with corresponding unit eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 0.707 \\ 0.707 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0.707 \\ -0.707 \\ 0 \end{pmatrix}.$$

The two in-plane principal directions are oriented at 45° and -45° relative to the x_1 axis.

Since this is a 2D problem, we can also calculate the principal strains using the formula

$$\varepsilon_{1,2} = \frac{1}{2}(\varepsilon_{11} + \varepsilon_{22}) \pm \sqrt{\left(\frac{\varepsilon_{11} - \varepsilon_{22}}{2}\right)^2 + \varepsilon_{12}^2} = \pm 0.002$$

which reproduces the two in-plane eigenvalues; the out-of-plane principal strain is zero.

Exercises 2.4:

1. Given the tensor field

$$\boldsymbol{\varepsilon} = \begin{pmatrix} 4kX_1 & 3kX_2 & 0 \\ 3kX_2 & 2kX_1 & 0 \\ 0 & 0 & 2kX_3 \end{pmatrix}$$

where $k = 10^{-4}$, determine the principal strains and directions at $\mathbf{X} = (1, 1, 1)$. Compare the two in-plane eigenvalues with the values obtained from the 2D formula.

2. Given the displacement field

$$\mathbf{u} = k(X_1^2 + X_2^2)t \hat{\mathbf{e}}_1 + kX_1X_2\hat{\mathbf{e}}_2 + kX_1X_3\hat{\mathbf{e}}_3$$

where $k = 10^{-4}$, determine the principal strains and directions at point $\mathbf{X} = (2, 1, 1)$ for $t = 1$ and $t = 2$.

3. Given the infinitesimal strain tensor $\boldsymbol{\varepsilon}$, find the principal strains and directions. Consider $k = 10^{-4}$.

$$\boldsymbol{\varepsilon} = k \begin{pmatrix} 1 & 0.5 & 0 \\ 0.5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2.15 Infinitesimal strain in cylindrical coordinates

In cylindrical coordinates (r, θ, z) , the displacement field is

$$\mathbf{u} = u_r \hat{\mathbf{e}}_r + u_\theta \hat{\mathbf{e}}_\theta + u_z \hat{\mathbf{e}}_z \quad (2.60)$$

where $\hat{\mathbf{e}}_r$, $\hat{\mathbf{e}}_\theta$, and $\hat{\mathbf{e}}_z$ are unit vectors in the radial, circumferential, and axial directions, respectively. [Figure 2.6](#) shows the strain components in an infinitesimal volume.

The physical components of the infinitesimal strain tensor are

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{rr} & \varepsilon_{r\theta} & \varepsilon_{rz} \\ \varepsilon_{\theta r} & \varepsilon_{\theta\theta} & \varepsilon_{\theta z} \\ \varepsilon_{zr} & \varepsilon_{z\theta} & \varepsilon_{zz} \end{pmatrix} \quad (2.61)$$

where each component is given by

$$\begin{aligned} \varepsilon_{rr} &= \frac{\partial u_r}{\partial r} \\ \varepsilon_{\theta\theta} &= \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} \\ \varepsilon_{zz} &= \frac{\partial u_z}{\partial z} \\ \varepsilon_{r\theta} &= \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right) \\ \varepsilon_{rz} &= \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right) \\ \varepsilon_{\theta z} &= \frac{1}{2} \left(\frac{\partial u_\theta}{\partial z} + \frac{1}{r} \frac{\partial u_z}{\partial \theta} \right) \end{aligned} \quad (2.62)$$

Infinitesimal strain in cylindrical coordinates

For an axisymmetric deformation without circumferential displacement, $u_\theta = 0$ and all derivatives with respect to θ vanish. The components reduce to

$$\begin{aligned} \varepsilon_{rr} &= \frac{\partial u_r}{\partial r} & \varepsilon_{\theta\theta} &= \frac{u_r}{r} & \varepsilon_{zz} &= \frac{\partial u_z}{\partial z} \\ \varepsilon_{r\theta} &= 0 & \varepsilon_{rz} &= \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right) & \varepsilon_{\theta z} &= 0 \end{aligned} \quad (2.63)$$

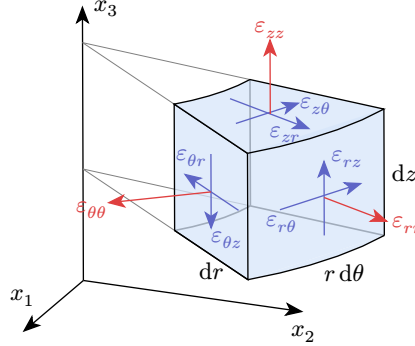


Figure 2.6: Components of the infinitesimal strain tensor in cylindrical coordinates.

Example 2.9: The Cartesian components of ε below are given at the point $(X_1, X_2, X_3) = (4, -3, 2)$. Determine its cylindrical components.

$$\varepsilon = \begin{pmatrix} 0.0136 & 0.0048 & -0.008 \\ 0.0048 & 0.0164 & 0.006 \\ -0.008 & 0.006 & 0.005 \end{pmatrix}$$

Solution:

Changing from Cartesian to cylindrical components is an orthogonal basis transformation. The cylindrical unit vectors, expressed in the Cartesian basis, are

$$\hat{e}_r = \cos(\theta)\hat{e}_1 + \sin(\theta)\hat{e}_2, \quad \hat{e}_\theta = -\sin(\theta)\hat{e}_1 + \cos(\theta)\hat{e}_2, \quad \hat{e}_z = \hat{e}_3$$

where $r = \sqrt{X_1^2 + X_2^2}$, $\cos(\theta) = \frac{X_1}{r}$, and $\sin(\theta) = \frac{X_2}{r}$.

Then, the transformation tensor is

$$\mathbf{R} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

At $(4, -3, 2)$, $r = 5$, $\cos(\theta) = \frac{4}{5}$, and $\sin(\theta) = -\frac{3}{5}$. Therefore,

$$\mathbf{R} = \begin{pmatrix} 0.8 & 0.6 & 0 \\ -0.6 & 0.8 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The cylindrical components are

$$\boldsymbol{\varepsilon}' = \mathbf{R}^T \boldsymbol{\varepsilon} \mathbf{R} = \begin{pmatrix} 0.01 & 0 & -0.01 \\ 0 & 0.02 & 0 \\ -0.01 & 0 & 0.005 \end{pmatrix}$$

Exercises 2.5:

1. The cylindrical components $\boldsymbol{\varepsilon}'$ were calculated at $(r, \theta, z) = (3, \frac{\pi}{6}, 0)$. Convert them to Cartesian components.

$$\boldsymbol{\varepsilon}' = \begin{pmatrix} 0.002 & -0.001 & 0.002 \\ -0.001 & 0.001 & 0 \\ 0.002 & 0 & 0.003 \end{pmatrix}$$

2.16 Infinitesimal strain in spherical coordinates

Use spherical coordinates (ρ, θ, φ) , where ρ is the radial coordinate, θ is the polar angle measured from the positive x_3 axis, and φ is the azimuthal angle in the x_1 - x_2 plane. The displacement field is

$$\mathbf{u} = u_\rho \hat{\mathbf{e}}_\rho + u_\theta \hat{\mathbf{e}}_\theta + u_\varphi \hat{\mathbf{e}}_\varphi, \quad (2.64)$$

where $\hat{\mathbf{e}}_\rho$, $\hat{\mathbf{e}}_\theta$, and $\hat{\mathbf{e}}_\varphi$ are the associated orthonormal basis vectors. The physical components of the infinitesimal strain tensor are

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{\rho\rho} & \varepsilon_{\rho\theta} & \varepsilon_{\rho\varphi} \\ \varepsilon_{\theta\rho} & \varepsilon_{\theta\theta} & \varepsilon_{\theta\varphi} \\ \varepsilon_{\varphi\rho} & \varepsilon_{\varphi\theta} & \varepsilon_{\varphi\varphi} \end{pmatrix} \quad (2.65)$$

where each component is given by

$$\begin{aligned} \varepsilon_{\rho\rho} &= \frac{\partial u_\rho}{\partial \rho} \\ \varepsilon_{\theta\theta} &= \frac{1}{\rho} \frac{\partial u_\theta}{\partial \theta} + \frac{u_\rho}{\rho} \\ \varepsilon_{\varphi\varphi} &= \frac{1}{\rho \sin(\theta)} \frac{\partial u_\varphi}{\partial \varphi} + \frac{u_\rho}{\rho} + \frac{u_\theta}{\rho} \cot(\theta) \\ \varepsilon_{\rho\theta} &= \frac{1}{2} \left(\frac{1}{\rho} \frac{\partial u_\rho}{\partial \theta} + \frac{\partial u_\theta}{\partial \rho} - \frac{u_\theta}{\rho} \right) \\ \varepsilon_{\rho\varphi} &= \frac{1}{2} \left(\frac{1}{\rho \sin(\theta)} \frac{\partial u_\rho}{\partial \varphi} + \frac{\partial u_\varphi}{\partial \rho} - \frac{u_\varphi}{\rho} \right) \\ \varepsilon_{\theta\varphi} &= \frac{1}{2} \left(\frac{1}{\rho \sin(\theta)} \frac{\partial u_\theta}{\partial \varphi} + \frac{1}{\rho} \frac{\partial u_\varphi}{\partial \theta} - \frac{u_\varphi}{\rho} \cot(\theta) \right) \end{aligned} \quad (2.66)$$